

Trajectory-Level Objective in StarPO vs. Previous Methods

Previous methods (e.g., PPO (Schulman et al., 2017), GRPO (Shao et al., 2024a)):

$$J_{\text{step}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} [R(x, y)] \quad (\text{optimize single-turn output } y \text{ given input } x)$$

StarPO (ours):

$$J_{\text{StarPO}}(\theta) = \mathbb{E}_{\mathcal{M}, \tau \sim \pi_{\theta}} [R(\tau)] \quad (\text{optimize total reward over trajectory } \tau = \{s_0, a_0, r_0, \dots, s_K\})$$